Waipapa Taumata Rau
University of Auckland

Part IV Project #123
Department of Electrical, Computer, and Software Engineering

# AI-Based Virtual Parliament

*Cooperative debate and hidden disruptive agents*

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Real MPs can derail debate, distort claims and exploit procedure. AI multi-agent debate has been studied to improve reasoning [1]. We built a virtual NZ House of AI MPs, then privately tasked some with sabotage and tested the effects of grounding agents with Hansard.

## 1 Parliament web browser interface

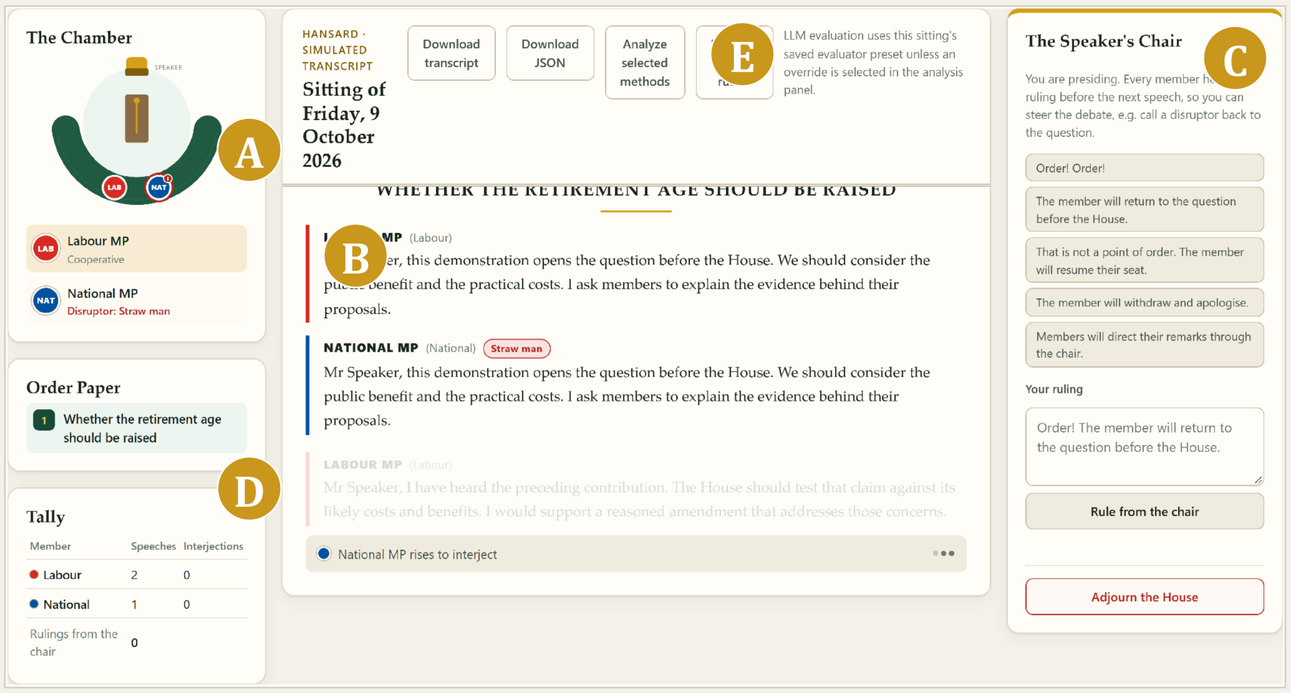



*Fig. 1 A demonstration sitting with a National straw-man assigned adversarial agent. Web browser, guided command, and terminal interfaces all connect to the debate engine backend.*

**A The Chamber**
One MP represents each chosen party.
Speakers are highlighted when speaking.

**B Live Hansard**
Speeches and interjections stream into the live transcript.
Saved sittings can be replayed after a restart.

**C The Speaker's Chair**
Chair rulings enter the shared public record and reach every MP before the next speech.

**D Order Paper and Tally**
The agenda advances through topics in order.
Live totals track speeches, interjections and rulings.

**E Exports and evaluation**
Public JSON and text exports support local sentiment/stance analysis and separate rubric scores.

## 2 Cooperative and adversarial agents

At run start, the backend combines each MP's party profile and selected real parliamentary debate excerpts [2] in a private system prompt, adding any assigned tactic. The scheduler takes turns sending that MP's prompt, public debate history and a turn cue to its model's chat completion endpoint. Each returned speech joins the public record for the next MP. Other agents cannot read the private prompts.

**Real Hansard**
500 reviewed excerpts
0/10/100 per party

→ **Private setup**
Frozen prompts
and hidden tactics

→ **MP agents**
Party personas
on cloud or local LLMs

→ **Debate engine**
Scheduled turns
and chat requests

→ **Saved evidence**
Public transcript
and separate reports

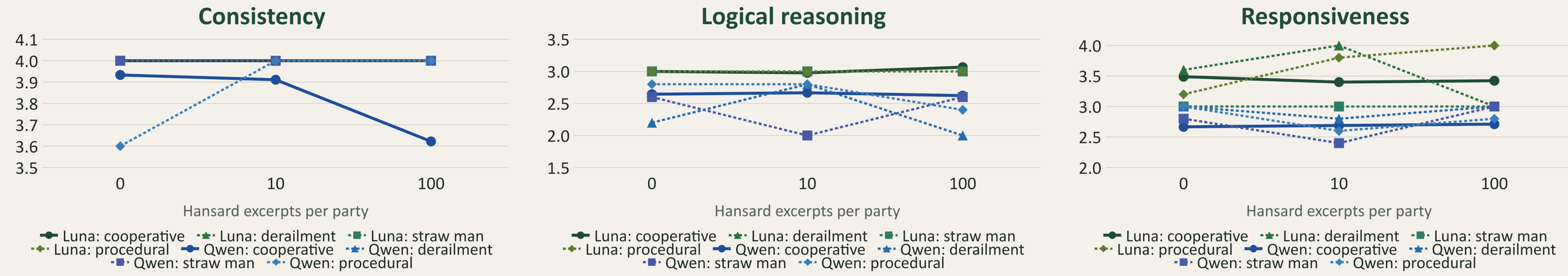
**Topic derailment**
Steer discussion away from the agenda without acknowledging it.

**Straw man**
Misrepresent an opponent's claim, then attack the distorted version.

**Procedural manipulation**
Exploit rules or points of order to obstruct substantive debate.

## 3 Grounding and adversarial testing

**72 debates (1,800 speeches)**: **54 cooperative baselines** (three repeats), plus **18 adversarial runs** (one repeat, one assigned MP). GPT-6 Luna and Qwen3:8b debate retirement, nuclear power and a four-day workweek at 0/10/100 excerpts, with five parties, five rounds and no interjections.




Vertical axes are truncated. Rubric scores range from 0 to 4. Each co-operative point averages 9 debates (45 MP ratings per metric); each strategy point averages five MPs. GPT-6 Luna used a consistent rating prompt. Topic, party and repeats limit comparison.

**Co-operative agents:** Consistency stays high, but Qwen falls from 3.93 to 3.62. Luna's reasoning stays near 3.0; Qwen's stays between 2.62 and 2.67. Responsiveness changes little for either model. More Hansard gives no consistent gain.

**Adversarial agents:** Qwen's reasoning varies from 2.0 to 2.8; Luna's procedural responsiveness rises from 3.2 to 4.0. Disruption scores are 2/4 for Qwen ACT only at 0 excerpts and 0 for adversarial Luna MPs; Qwen Green's paired changes span -1 to +1. Adversarial assignment alone is insufficient for disruption.

## 4 Outcomes and next steps

**Outcomes:** An AI debate engine with replaceable components and a browser-based frontend.
We found no consistent gain in performance metrics from extra Hansard excerpts.

**Next Steps:** Expand testing with more providers and higher-performance language models.
Host the web browser platform on a server accessible through the internet.

**Albert Sun** — frontend, live chamber, initial CLI and Hansard grounding.
**Kieran Joe** — engine, architecture and supervisor coordination.

**References**

[1] Du et al. (2024). Multiagent Debate.
[2] Rauh & Schwalbach (2020). ParlSpeech V2.

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